ALGEBRA I · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

6 question sets • 150 questions

Page numbers match the Question Collection.
Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • HSN-RN.A.1 • NUMBER & QUANTITY

Explain the meaning of rational exponents from the properties of integer exponents

MAIN	Algebra I main question	**HSN-RN.A.1**
BACKWARD 1	Apply the properties of integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Express the solution of an exponential equation as a logarithm	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship between exponents and logarithms	**HSF-BF.B.5**

SET 1 • HSN-RN.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

For the power-of-a-power property to hold, $(5^{1/3})^3$ must equal $5^{(1/3)\cdot 3} = 5$.

What number must $5^{1/3}$ be? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

5

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{5}$, the cube root of 5, because it is the number whose cube is 5.

GRADING NOTE Any equivalent form: $\sqrt[3]{5}$ or "the cube root of 5". The explanation must use the fact that $(5^{1/3})^3 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

Use the properties of exponents to explain why $8^{2/3} = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

6

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

$8^{2/3} = (8^{1/3})^2 = 2^2 = 4.$

GRADING NOTE Any equivalent reasoning, such as $8^{2/3} = (8^2)^{1/3} = \sqrt[3]{64} = 4$. The explanation must use a property of exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

For $x > 0$, which expression is equal to $x^{3/4}$?

A. the fourth root of x, cubed

B. $x^3 \div 4$

C. the cube root of x, to the fourth power

D. $3x \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

7

QUESTION ID

S1-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the fourth root of x, cubed

GRADING NOTE $x^{3/4} = (x^{1/4})^3$: the denominator 4 is the root and the numerator 3 is the power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

True or false?

$16^{1/2} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

8

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $16^{1/2}$ is $\sqrt{16} = 4$. An exponent of 1/2 is a square root, not a division by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

True or false?

For $x > 0$, $x^{1/2} \cdot x^{1/2} = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

9

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The product property gives $x^{1/2+1/2} = x^1 = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 1

Apply the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$2^3 \cdot 2^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

11

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2^8

GRADING NOTE Form required: a single power. Add exponents when multiplying powers with the same base: 3 + 5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

Write as a single power.

$(3^2)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

12

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3^8

GRADING NOTE Form required: a single power. Multiply exponents for a power of a power: 2 · 4 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

Write as a single power.

$5^6 \div 5^2$

A. 5^4

B. 1^4

C. 5^3

D. 5^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

13

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5^4

GRADING NOTE Subtract exponents when dividing powers with the same base: $6 - 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$(7^2)^3 = 7^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

14

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiply the exponents: $2 \cdot 3 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

$4^0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

15

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero number to the power 0 is 1, so $4^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

√81

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

17

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $9^2 = 81$, and √ means the nonnegative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

Find the value.

$\sqrt[3]{27}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

18

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

$\sqrt[3]{64}$

A. 32

B. 8

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

19

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $4^3 = 64$. (8 is $\sqrt{64}$, the square root.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

20

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-2)^3 = -8$, and a cube root of a negative number is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{25} = \pm 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

21

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The symbol √ means the nonnegative square root, so √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 1

Express the solution of an exponential equation as a logarithm • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $10^x = 500$. Write the solution as a logarithm, then use a calculator to round it to the nearest hundredth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

23

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log 500 \approx 2.70$

GRADING NOTE Form required: the logarithm and the rounded value. log means log base 10; log 500 = 2.69897...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

Solve $2^x = 32$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

24

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form ($\log_2 32 = 5$). $2^5 = 32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

Which expression is the solution of $3 \cdot 10^x = 60$?

A. $x = \log 20$

B. $x = \log 60$

C. $x = 20$

D. $x = \log 57$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

25

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x = \log 20$

GRADING NOTE Divide both sides by 3 first: $10^x = 20$, so $x = \log 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

True or false?

log 1,000 = 3

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

26

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^3 = 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = -10$ is $x = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

27

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10^x is positive for every real x, so $10^x = -10$ has no real solution. ($10^{-1} = 0.1$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 2

Use the inverse relationship between exponents and logarithms • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\log_2 8 = 3$ as an exponential equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

29

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2^3 = 8$

GRADING NOTE Form required: exponential form. log_b a = c means $b^c = a$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

$f(x) = 10^x$. Write the inverse function $f^{-1}(x)$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

30

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

log x (base 10), for x > 0

GRADING NOTE Any equivalent form: $\log_{10} x$. The inverse of 10^x is the base-10 logarithm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

What is the value of $\log_5 25$?

A. 5

B. 125

C. 2

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

31

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $5^2 = 25$, so $\log_5 25 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

True or false?

$10^{\log_{10} 7} = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

32

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10^x and log x are inverse functions, so each undoes the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log(10^x) = 10x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

33

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(10^x) = x$, because log undoes 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

SET 2 • HSN-RN.A.2 • NUMBER & QUANTITY

Rewrite expressions with radicals and rational exponents

MAIN	Algebra I main question	**HSN-RN.A.2**
BACKWARD 1	Solve equations of the form $x^2 = p$ and $x^3 = p$	**8.EE.A.2**
BACKWARD 2	Use the meaning of rational exponents	**HSN-RN.A.1**
FORWARD 1	Solve radical equations and identify extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Add, subtract, and multiply complex numbers	**HSN-CN.A.2**

SET 2 • HSN-RN.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Assume $x \geq 0$. Write $x^{5/2}$ using a radical.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

36

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(x^5)}$

GRADING NOTE Form required: radical form. $(\sqrt{x})^5$ is also correct. $x^{5/2} = (x^5)^{1/2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt[3]{(y^2)}$ using a rational exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

37

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y^{2/3}$

GRADING NOTE Form required: a rational exponent. The cube root gives the denominator 3, and the power 2 gives the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

Simplify.

$\sqrt{50} \cdot \sqrt{2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

38

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{50} \cdot \sqrt{2} = \sqrt{100} = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

Which expression is equal to $(27x^6)^{1/3}$ for every real number x?

A. $27x^2$

B. $3x^2$

C. $9x^2$

D. $3x^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

39

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $3x^2$

GRADING NOTE $27^{1/3} = 3$ and $(x^6)^{1/3} = x^2$ (a cube root is defined for every real number).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

True or false?

$\sqrt{(x^2)} = x$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

40

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$. For $x = -3$, $\sqrt{9} = 3$, not -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 1

Solve equations of the form $x^2 = p$ and $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^3 = 125$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

42

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5^3 = 125$, and $x^3 = 125$ has one real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

Solve for x. Give all solutions.

$x^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

43

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 6 and x = −6

GRADING NOTE Any equivalent form (x = ±6). Both solutions are required: $6^2 = (-6)^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

Solve for x.

$x^3 = -8$

A. ±2

B. −4

C. −2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

44

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −2

GRADING NOTE $(-2)^3 = -8$. A cube equation has one real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$x^2 = 7$ has the solutions $\sqrt{7}$ and $-\sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

45

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both $(\sqrt{7})^2$ and $(-\sqrt{7})^2$ equal 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

True or false?

$x^3 = 27$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

46

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only x = 3 works; $(-3)^3 = -27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 2

Use the meaning of rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$25^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

48

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $25^{1/2} = \sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B2-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

Find the value.

$27^{2/3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

49

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

Find the value.

$32^{1/5}$

A. 160

B. 2

C. 6.4

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

50

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE $2^5 = 32$, so the fifth root of 32 is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$9^{3/2} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

51

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$8^{1/3} = \frac{8}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

52

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $8^{1/3}$ is the cube root of 8, which is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 1

Solve radical equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 3) = 5. Check your solution.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

54

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. Square both sides: x + 3 = 25, so x = 22. Check: √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

Solve √(2x + 1) = x − 1. Check for extraneous solutions.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

55

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 (x = 0 is extraneous)

GRADING NOTE Form required: the solution, with x = 0 identified as extraneous. Squaring gives $x^2 - 4x = 0$, so x = 0 or 4. x = 0 gives √1 = 1 but 0 − 1 = −1, so it fails; x = 4 gives 3 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

What are the solutions of $\sqrt{x} = -3$?

A. $x = 3$

B. $x = 9$

C. $x = -9$

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

56

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. There is no solution.

GRADING NOTE $\sqrt{x}$ is never negative, so no x works. Squaring would give $x = 9$, but $\sqrt{9} = 3$, not -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

True or false?

Squaring both sides of an equation can introduce extraneous solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

57

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Squaring can make a false equation true, for example $-3 = 3$ becomes $9 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = -4$ is a solution of $\sqrt{(x + 8)} = x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

58

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{4} = 2$, but $-4 + 2 = -2$, so the equation is false at $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 2

Add, subtract, and multiply complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add.

(3 + 2i) + (1 − 5i)

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

60

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 − 3i

GRADING NOTE Form required: a + bi. Add real parts (3 + 1) and imaginary parts (2 − 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

Multiply.

$(2 + i)(3 - i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

61

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7 + i

GRADING NOTE Form required: a + bi. $6 - 2i + 3i - i^2 = 6 + i + 1 = 7 + i$, because $i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

Simplify.

$i^2 \cdot i$

A. $-i$

B. 1

C. i

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

62

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-i$

GRADING NOTE $i^2 = -1$, so $i^2 \cdot i = -i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

$(1 + i)^2 = 2i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

63

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

$i \cdot i = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

64

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i \cdot i = i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • HSN-RN.B.3 • NUMBER & QUANTITY

Explain why sums and products of rational and irrational numbers are rational or irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Find exact lengths and perimeters on the coordinate plane	**HSG-GPE.B.7**
FORWARD 2	Write numbers in the complex number system as a + bi	**HSN-CN.A.1**

SET 3 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why 5 + √2 is irrational. (Suppose 5 + √2 = r for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

67

QUESTION ID

S3-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

5 + √2 is irrational.

GRADING NOTE

REQUIRED WORK: An argument like this: if 5 + √2 = r with r rational, then √2 = r − 5, a difference of rational numbers, which is rational. But √2 is irrational, so 5 + √2 cannot be rational.

Any equivalent form of the argument. The key step is that rational numbers are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

Explain why $3\sqrt{7}$ is irrational. (Suppose $3\sqrt{7} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

68

QUESTION ID

S3-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$3\sqrt{7}$ is irrational.

GRADING NOTE

REQUIRED WORK: An argument like this: if $3\sqrt{7} = r$, then $\sqrt{7} = r \div 3$, a quotient of a rational number by a nonzero rational number, which is rational. But $\sqrt{7}$ is irrational, so $3\sqrt{7}$ is irrational.

Any equivalent form of the argument. It uses closure of the rational numbers under division by a nonzero number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

Explain why the sum of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

69

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

a/b + c/d = (ad + bc)/(bd). ad + bc and bd are integers and bd ≠ 0, so the sum is a quotient of integers.

GRADING NOTE Any equivalent explanation that writes the sum as a quotient of integers with a nonzero denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

Which number is irrational?

A. $0 \cdot \sqrt{5}$

B. $\sqrt{2} \cdot \sqrt{8}$

C. $\sqrt{3} + 2$

D. $\sqrt{9} + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

70

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $\sqrt{3} + 2$

GRADING NOTE $\sqrt{9} + 2 = 5$, $\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$, and $0 \cdot \sqrt{5} = 0$ are rational. $\sqrt{3}$ is irrational, so $\sqrt{3} + 2$ is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

True or false?

The sum of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

71

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} + (-\sqrt{2}) = 0$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √49 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

73

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Rational;

√49 = 7 = 7/1.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

The number 0.1010010001... continues with one more 0 before each 1. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

74

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Irrational; its decimal never ends and never repeats.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

Which number is rational?

A. √2

B. √10

C. π

D. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

75

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2727... (the digits 27 repeat)

GRADING NOTE A repeating decimal is rational: 0.2727... = 27/99 = 3/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

76

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any integer n can be written as n/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

√20 is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

77

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20 is not a perfect square, so √20 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write 1.25 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

79

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/4

GRADING NOTE Any equivalent form, such as 125/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

Write $-3\frac{1}{3}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

80

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−10/3

GRADING NOTE Any equivalent form, such as 10/(−3). $-3\frac{1}{3} = -(9/3 + 1/3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

Which quotient is equal to 0.6?

A. $\frac{3}{5}$

B. $\frac{6}{100}$

C. $\frac{5}{3}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

81

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3/5

GRADING NOTE $3 \div 5 = 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-7 = \frac{-7}{1}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

82

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every integer is a quotient of itself and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

83

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined, so 2/0 is not a number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 1

Find exact lengths and perimeters on the coordinate plane • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (3, 0), and (0, 3).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

85

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6 + 3√2 units

GRADING NOTE Form required: exact form. The legs are 3 and 3, and the hypotenuse is √18 = 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

A square has vertices (0, 0), (1, 1), (0, 2), and (−1, 1). Find its exact perimeter, and say whether it is rational or irrational.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

86

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4√2 units; irrational

GRADING NOTE Form required: exact form. Each side is $\sqrt{(1^2 + 1^2)} = \sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

How long is a diagonal of the rectangle with vertices (0, 0), (4, 0), (4, 3), and (0, 3)?

A. 7

B. $\sqrt{7}$

C. 25

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

87

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE $\sqrt{(4^2 + 3^2)} = \sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (1, 0), and (0, 1) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

88

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The perimeter is 2 + √2, a rational number plus an irrational number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

True or false?

The triangle with vertices (0, 0), (6, 0), and (0, 8) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

89

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sides are 6, 8, and 10, so the perimeter is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 2

Write numbers in the complex number system as a + bi • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{-49}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

91

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. $\sqrt{-49} = \sqrt{49} \cdot \sqrt{-1} = 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

Name the real part and the imaginary part of 4 − 3i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

92

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Real part 4; imaginary part −3

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

Which of these is a real number?

A. i

B. i^2

C. 2i

D. 1 + i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

93

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. i^2

GRADING NOTE $i^2 = -1$, which is real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

True or false?

Every real number is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

94

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

True or false?

$i^2 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

95

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE i is defined so that $i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-Q.A.1 • NUMBER & QUANTITY

Use units to guide multistep solutions and to choose scales for graphs

MAIN	Algebra I main question	**HSN-Q.A.1**
BACKWARD 1	Convert measurement units with ratio reasoning	**6.RP.A.3.d**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Use density in a model	**HSG-MG.A.2**
FORWARD 2	Solve problems with velocity vectors	**HSN-VM.A.3**

SET 4 • HSN-Q.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

A car goes 32 miles on each gallon of gas, and gas costs $3.60 per gallon.

Use units to find the cost of gas for a 400-mile trip.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

98

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

$45

GRADING NOTE Any equivalent form. 400 miles × (1 gallon ÷ 32 miles) × ($3.60 ÷ 1 gallon) = $45; the miles and gallons cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

Convert 72 kilometers per hour to meters per second.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

99

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 meters per second

GRADING NOTE Any equivalent form. 72 km/h × 1,000 m/km ÷ 3,600 s/h = 20 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

The formula d = rt gives distance. A rate r is in meters per second and a time t is in minutes.

What must be done before multiplying?

A. Divide r by t.

B. Convert t to seconds (or r to meters per minute).

C. Nothing; the units already match.

D. Convert d to minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

100

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Convert t to seconds (or r to meters per minute).

GRADING NOTE The time units must match: (m/s) × s = m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

A graph will show a town's population each year from 1990 to 2020. The populations range from 12,000 to 31,000.

Choose a scale for the vertical axis, and explain your choice.

Scale: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

101

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

For example, 0 to 35,000 in steps of 5,000, labeled "Population"; every value fits and the steps are equal.

GRADING NOTE Any equivalent form: any labeled scale with equal steps that includes 12,000 to 31,000 (starting at 10,000 instead of 0 is also correct if the break is shown or stated).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

True or false?

A speed of 5 miles per hour is about 7.3 feet per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

102

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 × 5,280 ÷ 3,600 = 7.33... feet per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 1

Convert measurement units with ratio reasoning • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many inches are in 3.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

104

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

42

GRADING NOTE Any equivalent form. 3.5 × 12 = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

How many kilograms are in 2,500 grams?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

105

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form. 2,500 ÷ 1,000 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

How many feet are in 4 yards?

A. 12

B. 48

C. 36

D. $\frac{4}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

106

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

GRADING NOTE 1 yard = 3 feet, so 4 yards = 12 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

True or false?

1 hour = 3,600 seconds

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

107

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 60 minutes × 60 seconds = 3,600 seconds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 km = 500 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

108

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 km = 1,000 m, so 5 km = 5,000 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A walker goes $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

110

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. (3/4) ÷ (1/4) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

Half a pound of cheese costs $3.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

111

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6

GRADING NOTE Any equivalent form. $3 ÷ (1/2) = $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\frac{2}{3}$ gallon of paint covers $\frac{1}{3}$ of a fence.

How many gallons cover the whole fence?

A. $\frac{1}{2}$

B. $\frac{2}{9}$

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

112

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE (2/3) ÷ (1/3) = 2 gallons per fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{6}$ hour is a speed of 3 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

113

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1/2) ÷ (1/6) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

True or false?

Painting $\frac{1}{4}$ of a wall in $\frac{1}{2}$ hour is a rate of $\frac{1}{8}$ wall per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

114

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (1/4) ÷ (1/2) = 1/2 wall per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 1

Use density in a model • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

Copper has a density of 8.96 grams per cubic centimeter.

What is the mass of a 5 cubic-centimeter piece of copper?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

116

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

44.8 grams

GRADING NOTE Any equivalent form. Mass = density × volume = 8.96 × 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

A county has 240,000 people living in 600 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

117

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

400 people per square mile

GRADING NOTE Any equivalent form, with units. 240,000 ÷ 600 = 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

A 2 cm by 3 cm by 4 cm block has a mass of 60 grams.

What is its density?

A. 0.4 gram per cubic centimeter

B. 2.5 grams per cubic centimeter

C. 60 grams

D. 24 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

118

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5 grams per cubic centimeter

GRADING NOTE The volume is 24 cubic centimeters, and 60 ÷ 24 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Population density is the number of people divided by the area they live on.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

119

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Density is a quantity per unit of area (or volume).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

An object with a volume of 10 cubic centimeters and a density of 3 grams per cubic centimeter has a mass of 3.3 grams.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

120

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass = 3 × 10 = 30 grams.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 2

Solve problems with velocity vectors • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A boat heads east at 12 km/h across a river whose current flows north at 5 km/h.

What is the boat's resulting speed?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5

122

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

13 km/h

GRADING NOTE Any equivalent form. The velocities are perpendicular, so the speed is $\sqrt{(12^2 + 5^2)} = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5

A plane points north at 300 km/h, and the wind blows west at 40 km/h.

Write its velocity as (east part, north part) in km/h.

Velocity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5

123

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(−40, 300)

GRADING NOTE Form required: (east part, north part). West is the negative east direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5

Two velocities, (3, 4) and (5, −1), are added.

What is the result?

A. (8, 5)

B. (15, −4)

C. (2, 5)

D. (8, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5

124

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (8, 3)

GRADING NOTE Add the parts: (3 + 5, 4 + (−1)) = (8, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5

True or false?

An object with velocity (6, 8) meters per second has a speed of 10 meters per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5

125

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{(6^2 + 8^2)} = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5

True or false?

Adding the velocities (3, 0) and (0, 4) gives a speed of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5

126

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is (3, 4), whose speed is √(9 + 16) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5

SET 5 • HSN-Q.A.2 • NUMBER & QUANTITY

Define appropriate quantities for descriptive modeling

MAIN	Algebra I main question	**HSN-Q.A.2**
BACKWARD 1	Use variables to represent quantities in a situation	**6.EE.B.6**
BACKWARD 2	Write and solve px + q = r from a word problem	**7.EE.B.4.a**
FORWARD 1	Use geometric quantities to make design decisions	**HSG-MG.A.3**
FORWARD 2	Recognize surveys, experiments, and observational studies and th…	**HSS-IC.B.3**

SET 5 • HSN-Q.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2

A school wants to model the total cost of a field trip by bus.

Name the quantities the model needs, with a unit for each.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

129

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

The number of students (students), the students each bus holds (students per bus), the cost of each bus (dollars per bus), and the admission price (dollars per student).

GRADING NOTE Any equivalent form: any complete, sensible set of quantities, each with a unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

A club rents a room for $300 and pays $40 for each member who attends. Its cost model is C = 40n + 300.

Define C and n, with units.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

130

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

C is the total cost in dollars; n is the number of members who attend.

GRADING NOTE Any equivalent wording, with both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

Which quantity best describes how fast a phone battery runs down?

A. Number of apps on the phone

B. Width of the screen in inches

C. Mass of the phone in grams

D. Percent of charge used per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

131

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Percent of charge used per hour

GRADING NOTE A rate of drain needs an amount of charge per unit of time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

Two towns have different populations and different areas.

Define one quantity, with units, that compares how crowded they are.

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

132

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

Population density: people per square mile (or per square kilometer).

GRADING NOTE Any equivalent quantity that is people per unit of area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

True or false?

Kilowatt-hours per month is a sensible quantity for a model of a household's electricity use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

133

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It measures energy used per unit of time, which is what an electricity bill is based on.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 1

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Tickets cost $8 each.

Write an expression for the cost of n tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

135

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A worker earns $15 per hour plus a $20 bonus.

Write an expression for the pay for h hours.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

136

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15h + 20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

A typist types 30 words per minute.

Which expression gives the number of words typed in m minutes?

A. 30m

B. $\frac{m}{30}$

C. m + 30

D. $\frac{30}{m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

137

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30m

GRADING NOTE Words = rate × time = 30m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

True or false?

If w is a number of weeks, 7w is the number of days.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

138

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each week has 7 days.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

True or false?

If d is a number of dollars, d + 5 is 5 times as many dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

139

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5 times as many is 5d; d + 5 is 5 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 2

Write and solve px + q = r from a word problem • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a $25 fee plus $10 per class. Tom paid $95.

How many classes did he take?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

141

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. 25 + 10c = 95 gives c = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

A phone costs $50 plus $15 per month. The total so far is $140.

For how many months has it been paid?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

142

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 50 + 15m = 140 gives m = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

A plumber charges a $12 fee plus r dollars per hour. A 3-hour job costs $57.

Which equation fits?

A. 3r + 12 = 57

B. 3(r + 12) = 57

C. 12r + 3 = 57

D. r + 3 + 12 = 57

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

143

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3r + 12 = 57

GRADING NOTE 3 hours at r dollars is 3r, plus the $12 fee.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

True or false?

x = 6 is the solution of 4x + 6 = 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

144

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4 \cdot 6 + 6 = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

145

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \cdot 3 - 5 = 1$; the solution is $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 1

Use geometric quantities to make design decisions • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden needs an area of at least 200 square feet and can use at most 60 feet of fence.

Does a 10 ft by 20 ft garden meet both requirements? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

147

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Its area is 200 square feet and its perimeter is 60 feet.

GRADING NOTE Both checks are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

A box with a square base 10 cm on each side must hold 1,000 cubic centimeters.

How tall must it be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

148

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 cm

GRADING NOTE Any equivalent form. 10 × 10 × h = 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

A rectangular pen must have an area of 100 square meters.

Which dimensions use the least fence?

A. 5 m by 20 m

B. 10 m by 10 m

C. 2 m by 50 m

D. 4 m by 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

149

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10 m by 10 m

GRADING NOTE The perimeters are 40, 50, 58, and 104 meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

True or false?

A 6 by 6 square has more area than a 4 by 8 rectangle, even though their perimeters are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

150

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 36 > 32, and both perimeters are 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with a perimeter of 20, the 1 by 9 rectangle has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

151

QUESTION ID
S5-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its area is 9; the 5 by 5 square has area 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 2

Recognize surveys, experiments, and observational studies and the variables they measure • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A researcher randomly assigns 40 plants to two fertilizers and measures each plant's growth in centimeters.

What kind of study is this, and what is the response variable (with units)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q

153

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

An experiment; the response is growth in centimeters.

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1

A survey asks 200 randomly chosen students how many hours they sleep on a school night.

What kind of study is this, and what variable does it measure?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q

154

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A survey (sample survey); hours of sleep per school night.

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2

Which kind of study can show cause and effect?

A. An observational study

B. A randomized experiment

C. A poll of volunteers

D. A survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q

155

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A randomized experiment

GRADING NOTE Only random assignment of treatments controls for other variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3

True or false?

In an observational study, the researchers do not assign treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q

156

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They observe groups that already exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4

True or false?

A survey of volunteers who answer an online poll is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q

157

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers choose themselves, so the sample is not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5

SET 6 • HSN-Q.A.3 • NUMBER & QUANTITY

Choose a level of accuracy that fits the limits of measurement

MAIN	Algebra I main question	**HSN-Q.A.3**
BACKWARD 1	Estimate to check whether an answer is reasonable	**7.EE.B.3**
BACKWARD 2	Round decimals to any place	**5.NBT.A.4**
FORWARD 1	Use volume formulas and report results to a sensible accuracy	**HSG-GMD.A.3**
FORWARD 2	Use the margin of error of a sample survey	**HSS-IC.B.4**

SET 6 • HSN-Q.A.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

A ruler measures to the nearest 0.1 cm. A rectangle measures 4.3 cm by 2.7 cm, and a calculator gives its area as 11.61 square cm.

Which report of the area best fits the accuracy of the measurements?

A. 11.610 square cm

B. 11.61 square cm

C. About 12 square cm

D. About 10 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

160

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. About 12 square cm

GRADING NOTE Each length could be off by 0.05 cm, so the true area is between about 11.3 and 12.0 square cm; the hundredths digit is not known.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

A stopwatch reads to the nearest 0.1 second. A runner covers 100 m in 12.4 seconds, and a calculator gives 8.0645161 meters per second.

Report the speed to a sensible accuracy.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

161

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 8.1 meters per second

GRADING NOTE Form required: a rounded value. 8.1 or 8.06 m/s is accepted; the true speed is between about 8.03 and 8.10 m/s, so more digits are not meaningful.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

A tape measure is marked in centimeters.

Which is the most sensible report of a person's height measured with it?

A. 167.3482 cm

B. 170.00 cm

C. 2 m

D. 167 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 167 cm

GRADING NOTE The tape measures to the nearest centimeter, so report whole centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

A bridge is 1,250 m long, measured to the nearest 10 m.

Between which two lengths is its true length?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

163

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

Between 1,245 m and 1,255 m

GRADING NOTE Any equivalent form. Measuring to the nearest 10 m allows an error of up to 5 m either way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

True or false?

Giving more decimal places always makes a measured answer more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

164

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A result cannot be more accurate than the measurements it comes from.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 1

Estimate to check whether an answer is reasonable • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate.

4.9×21.2

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

166

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

About 100

GRADING NOTE Any equivalent estimate near 100: $5 \times 20 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

Is 38.4 ÷ 4.8 = 80 reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

167

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 40 ÷ 5 = 8, so the answer should be about 8.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the best estimate of 19.8% of 503?

A. 50

B. 100

C. 1,000

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

168

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 100

GRADING NOTE About 20% of 500 is 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$29.95 × 4 is about $120.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

169

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $30 × 4 = $120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

True or false?

0.48 × 61 is about 300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

170

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE About 0.5 × 60 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 8.0645 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

172

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8.06

GRADING NOTE Form required: hundredths. The thousandths digit is 4, so round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

Round 11.61 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

173

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Form required: a whole number. The tenths digit is 6, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

Round 0.0357 to the nearest thousandth.

A. 0.035

B. 0.0357

C. 0.036

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

174

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.036

GRADING NOTE The ten-thousandths digit is 7, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

True or false?

3.149 rounded to the nearest tenth is 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

175

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The hundredths digit is 4, so round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

6.95 rounded to the nearest tenth is 6.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

176

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hundredths digit is 5, so round up to 7.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 1

Use volume formulas and report results to a sensible accuracy • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has a radius of 3.0 cm and a height of 10.0 cm.

Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

178

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

283 cubic centimeters

GRADING NOTE Form required: rounded to the nearest whole number. $V = \pi \cdot 3^2 \cdot 10 = 90\pi \approx 282.7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

A sphere has a radius of 5 m.

Find its volume exactly in terms of π, then to the nearest cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

179

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{500}{3}$π cubic meters, about 524 cubic meters

GRADING NOTE Form required: both the exact and the rounded value. V = (4/3)π · 125.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

A cone has a radius of 6 and a height of 4.

What is its volume?

A. 144π

B. 96π

C. 16π

D. 48π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

180

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 48π

GRADING NOTE $V = (1/3)\pi \cdot 6^2 \cdot 4 = 48\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

A cube with 2 cm edges has a volume of 8 cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

181

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a sphere doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

182

QUESTION ID
S6-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Volume depends on r^3, so it is multiplied by 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 2

Use the margin of error of a sample survey • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A poll reports that 54% of voters support a plan, with a margin of error of 3 percentage points.

What interval of values for the true percent does this give?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q

184

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

51% to 57%

GRADING NOTE Any equivalent form. 54% − 3% to 54% + 3%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1

A poll reports 54% support with a margin of error of 3 percentage points.

Do the results suggest that a majority supports the plan? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q

185

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Every value in the interval, 51% to 57%, is above 50%.

GRADING NOTE Any equivalent explanation that uses the interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2

A poll reports 48% support with a margin of error of 4 percentage points.

Which statement fits?

A. The poll must be wrong.

B. Fewer than half support the plan, for certain.

C. The true percent could be above or below 50%.

D. The true percent is exactly 48%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q

186

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The true percent could be above or below 50%.

GRADING NOTE The interval 44% to 52% includes values on both sides of 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3

True or false?

Larger random samples tend to have smaller margins of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q

187

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE More data gives less variation from sample to sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4

True or false?

A margin of error of 3 percentage points means the poll is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q

188

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives a range of plausible values, not the exact error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5